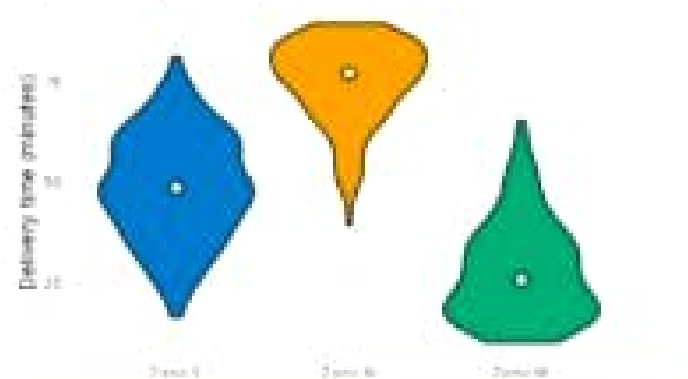


R.No: 4 Reg. No: 183624343 Name: B PALLAVI

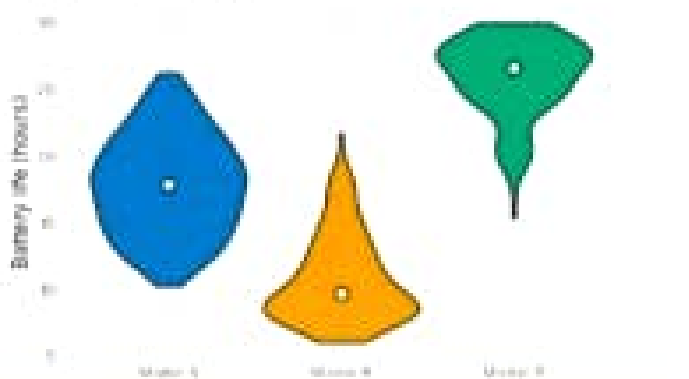
Part A (18 marks): Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Delivery time (minutes)



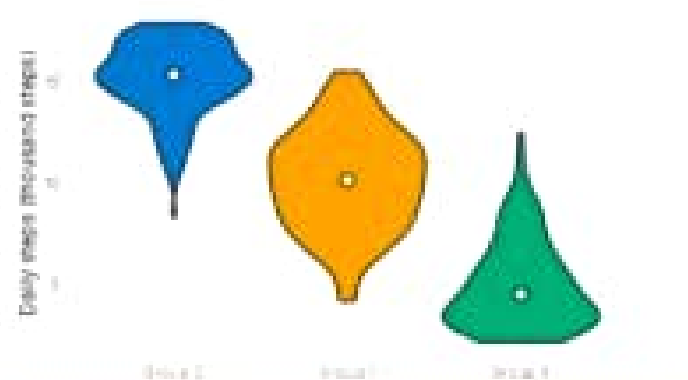
Violin	Skew (circle one)	Mean vs Median (circle)
Zone S	Right / Left / Symm	Mean > / < / = Median
Zone N	Right / Left / Symm	Mean > / < / = Median
Zone W	Right / Left / Symm	Mean > / < / = Median

Chart 2: Battery life (hours)



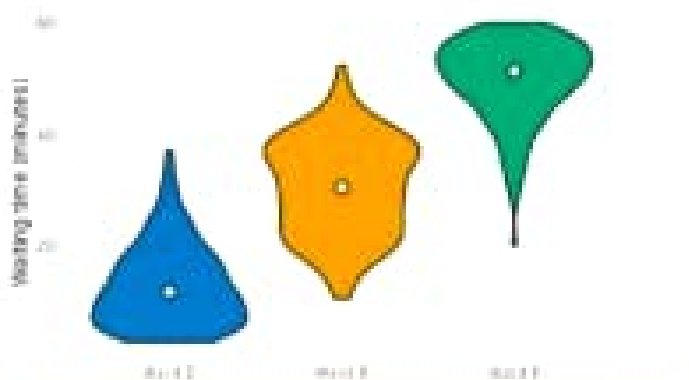
Violin	Skew (circle one)	Mean vs Median (circle)
Model S	Right / Left / Symm	Mean > / < / = Median
Model R	Right / Left / Symm	Mean > / < / = Median
Model P	Right / Left / Symm	Mean > / < / = Median

Chart 3: Daily steps (thousand steps)



Violin	Skew (circle one)	Mean vs Median (circle)
Group 2	Right / Left / Symm	Mean > / < / = Median
Group 3	Right / Left / Symm	Mean > / < / = Median
Group 4	Right / Left / Symm	Mean > / < / = Median

Chart 4: Waiting time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle)
Ward D	Right / Left / Symm	Mean > / < / = Median
Ward B	Right / Left / Symm	Mean > / < / = Median
Ward F	Right / Left / Symm	Mean > / < / = Median

Part B (2 marks)

- B1. In Chart 4, violin "Ward D" is not symmetric. Towards which side is its tail, and which is the better measure of a typical waiting time here: mean or median? Give one reason. [1]
- B2. Write one ggplot2 line that draws violins of value for each group from a data frame df [columns: group, value], with the median shown. [1]

8/10/2026

Activity

Chart 1 : Delivery Time.

* Zone S has a symmetric distribution, so the mean is approximately equal to the median.

* Zone N has a left-skewed distribution, so the mean is less than the median.

* Zone W has a right-skewed distribution, so the mean is less than the median.

Chart 2 : Battery life

* Model S has a symmetric distribution, so the mean is approximately equal to the median.

* Model R has a right-skewed distribution, so the mean is greater than the median.

* Model P has a left-skewed distribution, so the mean was less than the median.

Chart 3 : Daily steps.

* Group 2 has a left-skewed distribution, so the mean is less than the median.

* Group 1 has a symmetric distribution, so the mean is approximately equal to the median.

* Group 4 has a right-skewed distribution, so the mean is greater than the median.

chart 4: waiting Time

* ward D has a right-skewed distribution, so the mean is greater than the median.

* ward B has a left-skewed distribution, so the mean is less than the median.

* ward E has a left-skewed distribution, so the mean is less than the median.

B1.

ward D is right-skewed because its tails extends towards higher waiting times. The median is a better measure of typical waiting time because it is less affected by the long right tail.

B2.

```
ggplot(df, aes(x=group, y=Value)) + geom_violin() +  
Stat-summary(fun = median, geom = "point")
```